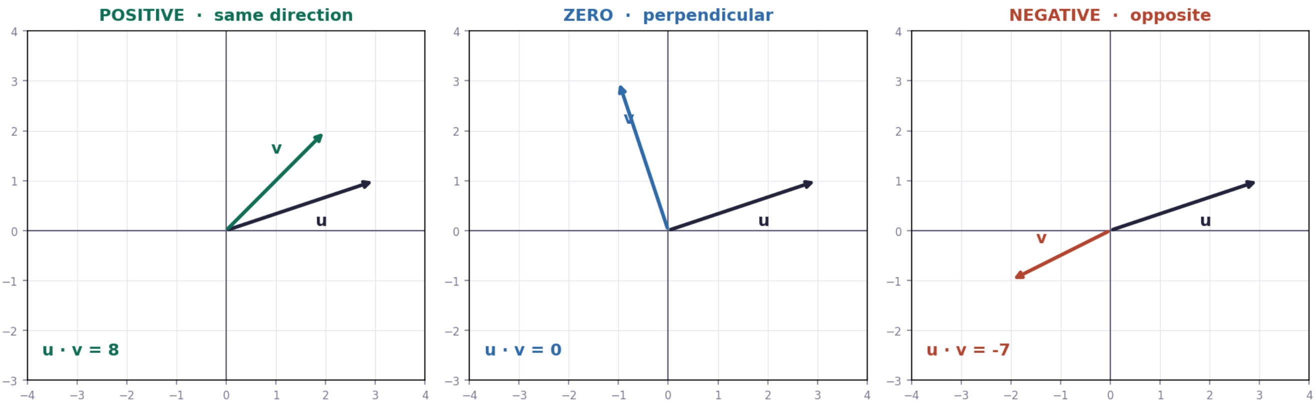
Dot products and duality

Two vectors in, one number out — and what that number tells you.

THE IDEA

The dot product multiplies matching components and adds them, giving a **single number**: (a, b) · (c, d) = ac + bd. Geometrically it measures how much two vectors point *the same way*.

Result	Meaning
positive	The vectors point in the same general direction
zero	They are **perpendicular** (90°)
negative	They point in generally opposite directions



DUALITY — THE SURPRISING PART

Taking the dot product with a fixed vector is the same thing as applying a **1×2 matrix** that squashes 2D down to a number. So every vector has a "twin" that acts as a transformation — that correspondence is what Grant calls **duality**.

CHAPTER 9 — WORKED EXAMPLES

THE FORMULA

(a, b) · (c, d) = ac + bd → the answer is ONE NUMBER

EXAMPLE 1

u = (3, 1)   v = (2, 2)

u · v = (3)(2) + (1)(2) = 6 + 2 = 8

positive → they point in the same general direction

EXAMPLE 2 — perpendicular

u = (3, 1)   v = (−1, 3)

u · v = (3)(−1) + (1)(3) = −3 + 3 = 0

zero → perpendicular (90°)

SIGN MEANING

> 0	same general direction
= 0	perpendicular
< 0	opposite directions

ML link: cosine similarity = dot product of two UNIT vectors.

Why it matters for ML: this is the most-used operation in machine learning. A neuron computes the dot product of inputs and weights. Similarity between two embeddings is **cosine similarity** — the dot product of the two *unit* vectors (Chapter 1's normalization).

TEST — Chapter 9

1. Compute (2, 5) · (3, 1). **→ 11**
2. Compute (4, −2) · (1, 2), and say what the sign means. **→ 0; the vectors are perpendicular**
3. Compute (−3, 1) · (2, −4). **→ −10 (generally opposite directions)**

FORMULAS FROM THIS CHAPTER

Dot product	(a,b)·(c,d) = ac + bd
Positive	same general direction
Zero	perpendicular
Negative	opposite directions
Cosine similarity	dot product of two unit vectors